Nithya Thushara Krishna Pothula

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Education

- **University of California** — Riverside, CA
  Master of Science in Computer Science; GPA: 3.59/4.00 — *Sep. 2023 – Mar. 2025*
- **MVJ College of Engineering** — Bengaluru, India
  Bachelor of Engineering in Computer Science; GPA: 8.1/10.0 — *Aug. 2017 – Jul. 2021*

Experience

- **T-Systems ICT India Pvt. Ltd** — Bengaluru, India
  Associate Consultant – Cyber Recovery, Agile, ServiceNow, ITIL — *Dec. 2021 - Sep. 2023*
  - **Cyber Recovery Execution & Risk Mitigation**: Led Cyber Recovery initiatives to safeguard enterprise workloads, ensuring compliance with SLAs and optimizing cyber resilience. Been part of automated recovery frameworks, improving recovery efficiency.
  - **Project Management & Process Automation**: Managed end-to-end onboarding and evergreen programs, optimizing enterprise Cyber Recovery strategies. Utilized Agile methodologies, JIRA, and ServiceNow to streamline project execution and automate ITSM processes.
  - **ITIL-Compliant Incident & Request Management**: Oversaw ITIL-aligned Problem and Request Management, ensuring process efficiency and risk mitigation. Conducted quality assurance audits and documentation reviews, reducing compliance gaps by 25%.
  - **Trusted Cybersecurity Advisor**: Provided expert consultation on Cyber Recovery strategies, helping global customers enhance their business continuity and security resilience against cyber threats.
- **Center for Artificial Intelligence & Robotics (CAIR), DRDO** — Bangalore, India
  Project Trainee – Deep Learning, IoT, TensorFlow, OpenCV — *Mar. 2021 - Jul. 2021*
  - **Face Mask and Temperature Detection System**: Developed and deployed a deep learning-based contactless screening system using CNN (MobileNetV2) and IoT sensors. Achieved a 95% accuracy in real-time face mask classification.
  - **IoT Integration and Deployment**: Integrated MLX90614 infrared temperature sensors and ultrasonic sensors for real-time user detection, reducing manual monitoring efforts by 40%.
  - **Optimized Model Performance**: Enhanced system reliability by applying image preprocessing, dataset augmentation, and hyperparameter tuning, reducing false positives by 20%. Deployed on Raspberry Pi for real-world implementation.
- **Center for Artificial Intelligence & Robotics (CAIR), DRDO** — Bangalore, India
  Project Trainee – Robot Operating System (ROS), C++ — *Jul. 2019 - Aug. 2019*
  - **Prohibition Layer Plugin for Robot Navigation**: Developed a Prohibition Layer Plugin in costmap_2D to restrict robot movement in predefined areas, improving navigation safety for mission-critical defense applications.
  - **Path Planning and Obstacle Avoidance**: Integrated ROS Navigation Stack with pathfinding algorithms using C++, optimizing route efficiency and enhancing obstacle avoidance by 35%.
  - **Dynamic Configuration and Modularity**: Configured layered costmaps dynamically via YAML files, enabling real-time navigation adjustments. Utilized pluginlib for modular development, ensuring seamless integration into ROS-based robotic systems.

Academic Projects

- **Graph Partitioning for Spatial Regionalization**: Implemented PRRP-based graph partitioning, ensuring 100% spatial contiguity in regionalization. Optimized partitioning algorithms, achieving a 4x speedup over existing techniques. Reproduced SIGSPATIAL 2023 results and validated statistical significance in regionalization.
- **BERT-Enhanced Information Retrieval**: Built a hybrid search system using BERT embeddings and Lucene, improving search precision by 35%. Optimized FAISS-based indexing, reducing query response time from 1.5s to 0.7s. Indexed 500K+ documents, enhancing retrieval efficiency and ranking relevance by 40%. Designed a Flask-based API for real-time search with 98% uptime. Reduced tokenization overhead by 50%, improving embedding efficiency for 10K+ word documents.
- **MobileInsight for Network Analysis**: Analyzed mobile network traces using MobileInsight in a virtual environment, extracting key user and performance metrics. Developed protocol replay, log filtering, and uplink latency analysis scripts for efficient network evaluation. Configured QCSuper for 2G/3G/4G/5G radio frame analysis, enhancing low-level monitoring. Optimized log filtering, reducing dataset size and improving computational efficiency.
- **Image Compression using CUDA Libraries**: Developed a GPU-accelerated image compression model using CUDA, cuFFT, and cuBLAS for real-time processing. Implemented parallelized Wavelet Transform, Quantization, and DCT compression, reducing image size by 90%. Optimized Run-Length Encoding (RLE) with CUDA kernels, improving compression efficiency. Integrated OpenCV for image preprocessing and evaluation, ensuring seamless workflows.

Skills

C/C++, Java, Python, Data Structures & Algorithms, SQL, PHP, Web Development, HTML/CSS, MongoDB, Windows/Linux, Git, Jira, ServiceNow, Agile, Team Management, Client Relations.